DRAFT — Natural Gas Sector Framework: Design Basis and Military Action Extension

Design Basis and Military Action Extension

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Preface

The natural gas transmission system is one of the most consequential and least formally analyzed infrastructure systems in the United States. Natural gas powers approximately 40% of U.S. electricity generation, heats more than 50 million American homes, and fuels the industrial processes on which the domestic economy depends. Yet the sector operates without a formal Design Basis — without a documented, methodologically rigorous statement of the bounding events its facilities must be designed to survive. Pipeline safety regulations address corrosion and mechanical damage. Emergency planning addresses extreme weather. But no unified framework asks the foundational engineering question: what is the worst credible event this system must withstand, and what does adequate response to that event require?

This document fills that gap and immediately extends it. The first contribution of the DBA-GAS sector framework is to establish a Design Basis for the natural gas sector — a structured methodology for identifying bounding events, defining consequence envelopes, and specifying the system response standards that constitute adequate protection. The second contribution is to extend that basis to the military-action threat tier: to ask what happens when the bounding event is not an equipment failure, a natural hazard, or an operator error, but a deliberate, sophisticated, state-directed attack. Those two contributions are inseparable. A Design Basis that does not address intentional action is incomplete for a sector that the federal government has formally designated as critical to national security.

The DBA-MA framework was developed in direct response to the demonstrated reality of military action against energy infrastructure. Nord Stream I and II proved that state actors will destroy major pipeline infrastructure as an instrument of coercive strategy. The Russian campaign against Ukrainian energy infrastructure — systematically targeting gas compression, storage, and generation through two consecutive winter seasons — proved that sustained energy infrastructure degradation is an operational doctrine, not an improvised tactic. The Colonial Pipeline ransomware incident proved that even a criminal actor, not a state, can produce national-level consequences when the sector lacks mandatory security standards.

The author spent thirty years in critical infrastructure security, including service as Vice President and Chief Security Officer at the North American Electric Reliability Corporation, where the mandatory physical security standard CIP-014 was developed and implemented. The DBA-GAS framework applies the same foundational logic to the natural gas sector: define the bounding threat, establish what constitutes adequate protection, and provide the analytical foundation for building enforceable standards.

Seven visits to Ukraine and the February 2022 letter warning of imminent Russian military action against Ukrainian energy infrastructure inform the analysis of military action in this document. The threat is not theoretical. The design basis is drawn from events that have already occurred.

1. Introduction

1.1 Purpose

This document establishes the Design Basis and Design Basis Accident — Military Action (DBA-MA) framework for the natural gas sector. It serves two related purposes.

First, it introduces the Design Basis concept to the natural gas sector. It establishes the methodology for identifying and characterizing bounding events — the events that natural gas sector facilities and systems must be designed to withstand. This is foundational analytical work that the sector currently lacks. The non-adversarial Design Basis defined in Section 4 establishes the baseline resilience posture against which all subsequent analysis is measured.

Second, it extends that Design Basis to the military-action threat tier. Section 5 introduces the four characteristics of military action that distinguish it from the non-adversarial bounding events in Section 4 and establishes why those characteristics require an extended analytical framework. The DBA-MA extension is not a separate methodology layered onto the Design Basis — it is a natural and necessary development of it, prompted by the demonstrated reality of state-directed attacks against energy infrastructure.

1.2 Scope and Applicability

This document governs the DBA-GAS series. It applies to natural gas sector infrastructure, including upstream production and gathering, midstream transmission pipelines and compressor stations, natural gas processing, underground storage, local distribution, and LNG terminal infrastructure. The DBA-GAS series primarily addresses the U.S. natural gas sector, with reference to allied and international contexts in which the supply chain implications of military action extend beyond domestic boundaries.

Petroleum refining, crude oil pipelines, and petroleum product distribution are addressed in the DBA-OIL series. The electricity sector is addressed in a separate DBA-MA child document. Where natural gas and electricity infrastructure intersect — particularly in the gas-fired generation fleet and the gas-electricity interdependency loop — this document addresses the consequence chain as it propagates from the gas sector into the electricity sector, without displacing the electricity sector’s own analytical framework.

1.3 Relationship to Governing Documents

This document is issued under the DBA-MA Parent Framework (IAEA L1 v2.3). The parent framework establishes the governing threat taxonomy, consequence analysis methodology, and framework architecture. This sector framework inherits from the parent and adapts those foundations to the natural gas sector's specific characteristics.

Facility-class documents, use cases, and policy companions in the DBA-GAS series are issued under this sector framework. They inherit the threat taxonomy, bounding methodology, and analytical structure established here without repeating it. DBA-GAS-FC1 (Natural Gas Transmission), DBA-GAS-FC2 (Natural Gas Processing), DBA-GAS-FC3 (Underground Storage), DBA-GAS-FC4 (Local Distribution), DBA-GAS-UC-LNG (LNG Terminal Attack), and DBA-GAS-POL (Deterrence Without Standards) are currently issued under this framework.

DBA-EN series instruments are cited in this document by designation, without version tokens. The current version of each is stated in DBA-EN-REG, which is the authoritative register for this series; a version-bearing citation would duplicate that register and drift from it between revisions. Resolve any series instrument named in this document against DBA-EN-REG. External regulatory references and supporting documents retain their version and date tokens.

Command Broker family instruments are cited in this document by designation, without version tokens. The current version of each is stated in CB-REG, which is the authoritative register for this family.

1.4 How to Use This Document

The Design Basis concept originated in nuclear safety engineering in the 1960s and 1970s. Readers embedded in that tradition will recognize the framework immediately. Readers from the natural gas sector, policy community, or allied defense planning community may not. This section is written for the latter.

A Design Basis answers a specific engineering question: what is the worst credible event this facility or system must be built to survive? Not the worst imaginable event — the worst credible one. The distinction matters because design is a resource-constrained activity. Every facility cannot be hardened against every conceivable threat. A Design Basis defines the envelope: inside it, protection is mandatory; outside it, residual risk is accepted. Once the envelope is defined, the engineering question becomes tractable: which system design, operating procedures, and recovery architecture constitute adequate protection against the bounding event?

The Design Basis Accident — Military Action extension asks a harder version of the same question: what is the worst credible event when the initiating condition is deliberate military action rather than equipment failure or natural hazard? The answer is harder because an adversary adapts. The analysis in this document provides the tools to answer it.

Readers primarily interested in the military action analysis may proceed directly to Section 5 after reading Section 3. Readers who are primarily interested in the Design Basis foundation should read Sections 3 and 4 before proceeding. Readers responsible for facility-class document development should read the full document, as the sector framework establishes the analytical inheritance that all facility-class documents carry.

2. Sector Profile

2.1 Sector Structure

The natural gas sector is organized into five operational layers. Each layer has distinct physical characteristics, regulatory oversight, and consequence profiles under both non-adversarial and military-action conditions. Understanding the operational architecture is a prerequisite to defining the Design Basis.

2.1.1 Upstream: Production and Gathering

Upstream infrastructure includes producing wells, wellheads, gathering pipelines, field compression, and gas-liquid separators. Production is geographically dispersed across major producing basins — the Permian, Appalachian, Haynesville, Anadarko, and others. Individual well or gathering system failures have limited systemic consequences because the upstream layer benefits from inherent redundancy through geographic distribution. The upstream layer becomes analytically significant primarily at the basin level, where sustained degradation of multiple production nodes reduces the gas volume available to the transmission system.

2.1.2 Midstream: Transmission and Processing

Midstream infrastructure is the operationally critical and most consequential layer. It includes approximately 300,000 miles of interstate and intrastate natural gas transmission pipeline, operating at pressures typically between 200 and 1,500 psi. Compressor stations sustain these pipelines, spaced 50 to 100 miles apart; their failure collapses line pressure and interrupts supply.

Gas processing plants separate natural gas liquids (NGLs) from the raw gas stream. Processing is required before gas enters the transmission system in most producing regions; processing plant outages can curtail upstream production even when transmission pipelines are intact. Underground storage — depleted reservoirs and salt cavern fields — provides the buffering capacity that allows the transmission system to match variable demand, particularly during peak periods of heating and generation.

2.1.3 LNG Terminals

LNG terminals perform either liquefaction (export) or regasification (import). Export terminals cool natural gas to approximately −162°C, reducing volume by a factor of approximately 600 to enable ocean transport. Import terminals, floating storage and regasification units (FSRUs), and regasification units warm LNG back to a gaseous state for injection into the transmission system. LNG terminals are among the most capital-intensive facilities in the energy sector and occupy a strategic position in the allied energy supply chain: U.S. export terminals are the primary replacement supply source for European nations that shifted away from Russian pipeline gas after 2022.

2.1.4 Distribution

Local distribution companies (LDCs) receive gas at city gate stations from the transmission system and deliver it through lower-pressure distribution pipelines to end users: residential heating and cooking, commercial operations, and gas-fired generation. Distribution infrastructure is dense and geographically extensive, with millions of service connections. It operates under state public utility commission oversight rather than federal FERC jurisdiction.

2.1.5 Markets and Control Systems

Natural gas markets — the Henry Hub benchmark, regional trading hubs, forward markets, and physical balancing mechanisms — coordinate supply and demand across the sector. Market disruption under military action can compound physical supply disruptions, delaying recovery. Control systems — SCADA and distributed control systems governing pipeline operations, compressor management, storage withdrawal, and safety instrumented systems — are both the sector's operational nervous system and a primary target for cyber-kinetic combined-arms attacks.

2.2 Regulatory Authority

The natural gas sector operates under a fragmented regulatory structure in which safety jurisdiction, economic regulation, security oversight, and emergency response authority are distributed across multiple agencies without a unified framework. PHMSA holds pipeline safety jurisdiction under 49 CFR Parts 192 and 195; its integrity management programs address corrosion, mechanical damage, and natural hazards, but not the adversary-action threat tier. FERC holds economic and reliability jurisdiction over interstate pipelines and LNG terminals under the Natural Gas Act. TSA holds surface transportation security authority and has issued mandatory cybersecurity directives following the Colonial Pipeline incident; physical security requirements remain voluntary. CISA provides sector risk management support with advisory authority only. The USCG holds maritime security jurisdiction over LNG waterfront facilities under MTSA, providing the strongest regulatory framework in the sector for the subset of infrastructure it covers.

The absence of mandatory physical security standards across the pipeline sector — the gap that distinguishes the natural gas regulatory environment from the electricity sector’s NERC CIP-014 framework — is both a security posture gap and a deterrence gap. It is addressed in the DBA-GAS-POL policy companion.

2.3 Strategic Significance

The natural gas sector’s strategic significance derives from three interdependency arguments that elevate a supply disruption from an energy event to a national security event.

Electric power generation dependency. Natural gas accounts for approximately forty percent of U.S. electricity generation. In cold-weather regions, gas-fired generation provides the marginal capacity needed to maintain grid stability during peak demand. A sustained gas supply disruption does not merely inconvenience gas customers — it triggers electricity-generation shortfalls that cascade across every sector dependent on electricity.

Residential thermal energy dependency. More than fifty million American homes are heated primarily by natural gas. A supply disruption during the winter heating season creates life-safety consequences — hypothermia risk and hospital system overload — that compound the economic and cross-sector impacts of the initiating event. An adversary planning a winter attack deliberately exploits this vulnerability.

Allied supply chain dependency. U.S. LNG export capacity has become the primary replacement supply for European nations that eliminated their dependence on Russian pipeline gas after 2022. A sustained disruption of U.S. LNG export infrastructure does not merely affect domestic supply — it degrades allied energy security, strains NATO cohesion. It gives the adversary strategic leverage over European member-state governments managing heating-fuel shortages while simultaneously meeting defense obligations.

3. Design Basis: Natural Gas Sector

3.1 The Concept and Its Origins

The Design Basis concept was formalized in nuclear safety engineering in the 1960s and 1970s as the foundation of reactor safety analysis. The nuclear industry faced a specific analytical challenge: reactors could fail in many ways, but design and licensing resources were finite. A methodology was needed to identify the bounding cases — the worst credible events that safety systems must be designed to withstand — and to distinguish them from less severe events that existing safety margins would absorb without specific design accommodations.

The result was the Design Basis Accident: a postulated accident that represents the most severe conditions that safety systems must be capable of mitigating. The DBA is not the worst imaginable accident. It is the worst accident that can credibly result from defined equipment failures, operator errors, or natural hazards, given the facility’s design and operating envelope. Once the DBA is defined, the safety system design requirement is that the system must be capable of responding to the DBA without exceeding the defined consequence limits.

The concept has proven durable and analytically productive across six decades of nuclear safety practice. Its extension to other critical infrastructure sectors is not a metaphor — it is a direct application of the same analytical logic. The natural gas sector presents the same foundational challenge that nuclear engineering faced in the 1960s: a complex system with many possible failure modes, finite design and investment resources, and a need to identify the bounding cases that drive hardening and resilience requirements.

3.2 Application to the Natural Gas Sector

Applying the Design Basis methodology to the natural gas sector requires adapting the nuclear framework to a fundamentally different system architecture. Nuclear facilities are single-point systems: a reactor complex has defined boundaries, known safety systems, and a consequence chain that extends outward from a single facility. The natural gas sector is a networked system: consequence chains propagate through interconnected nodes — compressor stations, storage fields, processing plants, distribution networks — and the most significant events involve simultaneous or sequential failure of multiple nodes.

The Design Basis for the natural gas sector must therefore be defined at two levels. At the facility level, the Design Basis addresses the bounding events for individual asset classes: what is the worst credible event a compressor station must be designed to survive? At the system level, the Design Basis addresses the bounding events for the transmission network as an integrated system: what is the worst credible combination of facility-level events that the system must be capable of absorbing while maintaining critical supply functions?

Both levels are necessary. A facility-level Design Basis without a system-level counterpart produces facilities hardened against single-node events but a network unprepared for coordinated failure. A system-level Design Basis without facility-level grounding produces resilience planning that lacks the engineering specificity needed to drive hardening investment decisions.

The Design Basis serves a second function beyond establishing engineering requirements: it is a gap-finding instrument for policy, authority, and institutional coordination. Defining precisely what the sector must withstand makes visible the delta between that requirement and what existing regulatory frameworks actually compel. Every adequacy standard without a compliance mechanism, and every cross-sector coordination requirement without a pre-established protocol, is a gap that the Design Basis exposes—section 10 catalogs what this document reveals.

4. Design Basis Accidents: Non-Adversarial Envelope

This section constitutes the DBA layer — the second stage of the DB → DBA → DBA-MA sequence. The Design Basis in Section 3 defines the natural gas sector, explains why it is critical, and outlines the performance envelope. This section derives the specific postulated accident categories from that envelope and establishes the methodology and adequacy standards for analyzing each category. The DBA-MA extension in Section 5 then applies the four military action characteristics to each category. Non-adversarial bounding events fall into four categories: equipment failure, natural hazard, operational, and cascading failure. Each is addressed below with representative bounding events. The list is illustrative, not exhaustive; the framework will be updated as operational experience and analysis surface new candidates.

4.1 Bounding Event Methodology

A bounding event is identified through a structured process of hazard identification, consequence analysis, and credibility assessment. The methodology applied in this framework involves four steps.

Step 1 — Hazard identification. For each facility class and system layer, identify the universe of events that could initiate a consequence chain. This includes equipment failure modes, natural hazard exposures, operational errors, and — under the DBA-MA extension — adversary action modalities. Hazard identification is iterative: it is not completed once and fixed, but updated as new failure modes are observed, as threat intelligence evolves, and as facility-class documents develop more granular analyses.

Step 2 — Consequence analysis. For each identified hazard, trace the consequence chain: what does the initiating event cause, what systems respond or fail to respond, and what is the resulting consequence envelope? Consequence analysis must account for dependencies — the failure of one system that degrades another — and for recovery constraints that extend the duration of consequences beyond the initiating event.

Step 3 — Credibility assessment. Not every identified hazard represents a credible bounding event. The assessment asks: given the facility's design, location, operating envelope, and threat environment, is this event plausible? For non-adversarial events, credibility is assessed against historical frequency, engineering failure analyses, and natural-hazard exposure data. For adversarial events, credibility is assessed against the adversary's capability, demonstrated intent, and the target's strategic value.

Step 4 — Bounding selection. Among credible events, the bounding event is the one that produces the most severe consequence envelope given the defined assessment parameters. For facility-level analysis, the bounding event drives the design requirement. For system-level analysis, the bounding event defines the resilience architecture requirement.

The bounding events identified in this document and in facility-class documents are not exhaustive. They represent the analytical community's current best assessment of the most severe credible events. As the DBA-GAS series matures, as facility-class documents develop deeper analyses, and as operational experience and threat intelligence evolve, additional bounding events will be identified and incorporated. The framework is designed to accommodate that evolution: the four-step methodology is the stable foundation; the specific bounding events it produces are subject to revision.

4.2 Design Basis Adequacy Standards

Once the bounding event is defined, the Design Basis establishes what constitutes an adequate system response. For the natural gas sector, adequacy is defined against three criteria.

Supply continuity. The system must be capable of maintaining supply to priority customers — life-safety infrastructure, electric power generation, and firm-service residential and commercial customers — within defined time limits following the bounding event. The specific time limits

are defined at the facility-class level based on the consequence chain analysis for each bounding event.

Recovery capability. The system must possess the redundancy, mutual assistance capability, and equipment access needed to restore normal operations within a defined recovery window. Recovery capability requirements drive investment in spare equipment pre-positioning, portable compression capability, and storage infrastructure sizing.

Cross-sector consequence limitation. The system response must limit the propagation of the initiating event into dependent sectors — particularly the electricity sector — within defined bounds. Cross-sector consequence limits are established through the gas-electricity coordination framework described in Section 7.

4.3 Equipment Failure Events

4.3.1 Major Compressor Unit Failure

Initiating event: Failure of a primary compressor driver — gas turbine or reciprocating engine — at a major transmission compressor station, causing loss of compression capacity at that station. This is the most common high-consequence equipment failure mode in the transmission system. Major compressor units have long replacement lead times (12 to 24 months in normal supply chain conditions); the bounding case involves a failure mode that cannot be remediated through repair and requires unit replacement.

Consequence envelope: Line pressure downstream of the failed station drops progressively. Where the transmission system has parallel paths or alternative routing, the pressure loss can be partially offset. Where routing options are limited — as in constrained corridor configurations — downstream supply is curtailed. The bounding case assumes constrained routing and simultaneous peak demand conditions.

Adequacy standard: The system must maintain supply to priority customers through storage drawdown, alternative routing, or portable compression deployment for a period sufficient to procure and install replacement equipment. Planning assumptions for the adequacy standard are developed at the facility-class level.

4.3.2 Large-Diameter Pipeline Mechanical Failure

Initiating event: Failure of a large-diameter (24-inch or greater) high-pressure transmission pipeline segment from corrosion, third-party damage, or manufacturing defect, resulting in pipeline rupture and gas release. The bounding case involves a failure at a location where alternative routing is limited, and emergency response access is constrained.

Consequence envelope: Gas release and potential ignition at the rupture site. Pipeline segment isolated by block valve closure; supply to downstream customers curtailed during the isolation and repair period. The repair timeline is one to two weeks per rupture location for a cut-out and

weld repair; additional time for regulatory hydrotest and return to service. The bounding case involves locations where repair access is difficult — such as water crossings, urban environments, and mountainous terrain — thereby extending the repair timeline.

Adequacy standard: Block valve spacing and actuation capability must limit gas release volume and duration. Alternative supply routing or storage drawdown must sustain priority customer supply during the repair window. Environmental remediation capability must be pre-positioned for liquid-phase contamination at stream crossings.

4.3.3 Underground Storage Integrity Failure

Initiating event: Loss of well integrity at an underground storage field — depleted reservoir or salt cavern — resulting in uncontrolled gas migration, surface blowout, or loss of storage capacity. The Aliso Canyon storage blowout (2015–2016) is the anchor event for this bounding case: a single well integrity failure at a major storage field resulted in an 11-week blowout, the loss of approximately 5 billion cubic feet of gas, and supply curtailment for Southern California customers during a winter heating period.

Consequence envelope: Loss of storage buffer capacity during the outage period. If the event occurs at the onset of a high-demand period when storage withdrawal is relied upon to supplement transmission supply, the consequence chain extends from physical damage at the storage field to supply curtailment across the service territory. The bounding case involves a major storage field serving a region with limited alternative supply routing.

Adequacy standard: Storage field well integrity management programs must reduce the probability of integrity failure to acceptable levels. Alternative supply provisions — LNG peaking facilities, alternative storage access, and demand response protocols — must be capable of offsetting the loss of storage capacity for the duration of the repair and regulatory return-to-service process.

4.4 Natural Hazard Events

4.4.1 Polar Vortex Demand Surge

Initiating event: An extended polar vortex event producing extreme cold temperatures across a major consuming region, driving simultaneous peak demand for residential heating and electric power generation fuel supply. The February 2021 Winter Storm Uri event in Texas is the reference case: extreme cold caused simultaneous demand surges and supply curtailments due to wellhead freeze-offs and compressor station failures, resulting in cascading electric grid outages affecting millions of customers.

Consequence envelope: Demand exceeds the capacity of transmission and storage deliverability. LDCs initiate curtailment under priority protocols; interruptible customers — including gas-fired electric generators on interruptible contracts — are curtailed first. If curtailment does not restore

the supply-demand balance, firm-service customers face supply interruption. The feedback loop between gas curtailment and electric generation loss — gas compression requires electricity; electric generation requires gas — can produce cascading failures that exceed either sector's ability to recover independently.

Adequacy standard: Transmission and storage infrastructure must be capable of sustaining peak design demand conditions for a defined cold-weather event duration. Wellhead and compression infrastructure in producing basins must meet winterization standards appropriate to the region's design cold temperature. The gas-electric coordination framework must include pre-established curtailment sequencing that prevents the feedback loop from initiating.

4.4.2 Hurricane Disruption of Gulf Coast Production and Processing

Initiating event: A major hurricane (Category 3 or above) making landfall on or traversing the Gulf Coast, causing simultaneous disruption of offshore production platforms, onshore gathering and processing facilities, and coastal LNG terminals through physical damage, flooding, and extended power outages. Hurricanes Katrina and Rita (2005) are the anchor events: those storms simultaneously shut down approximately thirty percent of U.S. natural gas production and caused supply disruptions that extended weeks beyond the storms themselves.

Consequence envelope: Gulf Coast supply reduction propagates northward through the transmission system. Storage drawdown begins immediately; storage inventory depletion rate depends on the magnitude and duration of the supply shortfall and the coincidence with heating demand. If the storm makes landfall during the shoulder season (September–October), storage inventory is at or near its annual high and the buffer capacity is substantial. If the storm coincides with early winter demand, the buffer is smaller and the consequence envelope is larger.

Adequacy standard: Infrastructure in the Gulf Coast hurricane exposure zone must meet design standards for wind loading and storm surge appropriate to the design storm intensity. Storage inventory planning must account for the possibility of a major Gulf Coast supply disruption at the onset of the winter heating season. Alternative supply routing from non-Gulf-Coast producing regions must be available for emergency activation.

4.4.3 Seismic Event

Initiating event: A significant seismic event (magnitude 6.0 or greater on the Modified Mercalli scale) affecting a major producing basin, transmission corridor, or urban distribution network. The New Madrid Seismic Zone in the central United States presents the highest consequence scenario: a major New Madrid event could simultaneously damage transmission pipelines, compressor stations, storage fields, and the distribution network across a multi-state region with high gas dependency.

Consequence envelope: Physical damage to pipeline infrastructure in the event zone. Loss of pipeline pressure across the affected corridor. Distribution system damage — particularly at

older cast-iron and unprotected steel systems in urban areas — creates gas leak hazards requiring systematic shutdown and releak survey before restoration. The distribution restoration timeline following a major urban earthquake is measured in weeks to months, not days.

Adequacy standard: Transmission and distribution infrastructure in high-seismic-hazard zones must meet design standards for ground motions consistent with the region's design earthquake. Emergency response plans must address simultaneous multi-node damage scenarios and include pre-positioned repair resources for post-seismic restoration.

4.5 Operational Events

4.5.1 Control System Failure

Initiating event: Failure or loss of availability of SCADA systems governing transmission pipeline operations, resulting in loss of pipeline visibility and remote control capability. The bounding case involves a SCADA failure affecting a major pipeline system during peak demand, requiring a transition to manual operations across multiple compressor stations and block valve locations simultaneously.

Consequence envelope: Loss of real-time pipeline visibility degrades the operator’s ability to manage line pressure, respond to anomalies, and execute curtailment or isolation procedures. Manual operations increase response times and require additional field personnel. If the SCADA failure coincides with an initiating event elsewhere on the system — a compressor failure or pipeline anomaly — the degraded detection capability extends the consequence window of the secondary event.

Adequacy standard: SCADA systems must have redundant communications paths and backup control capability that maintains essential pipeline visibility and control during primary system failure. Operators must maintain manual operations capability and conduct periodic manual operations exercises. Recovery time objectives for SCADA restoration must be defined and tested.

4.5.2 Major Processing Plant Outage

Initiating event: Unplanned outage of a major natural gas processing plant in a producing region where processing is required before gas enters the transmission system, causing upstream production curtailment independent of transmission pipeline conditions. The bounding case involves a plant serving a significant share of a producing basin’s output during a period when storage inventory is constrained.

Consequence envelope: Gas volume entering the transmission system falls below the volume needed to maintain line pressure across downstream corridors. LDCs initiate demand curtailment. The nature of the plant damage governs the recovery timeline: a minor mechanical

failure may be remediable in days; a major process upset involving fired equipment may require weeks or months.

Adequacy standard: Processing plant redundancy requirements and outage notification protocols must be defined in coordination with transmission operators. Storage inventory planning must account for processing plant outage scenarios as a supply constraint in the design basis.

4.6 Cascading Failure Events

4.6.1 Gas-Electric Cascade

Initiating event: An event in either the natural gas or electricity sector that propagates into the other through the fuel supply and power consumption interdependency, creating a feedback loop that exceeds either sector's independent recovery capability.

Consequence envelope: The gas-electric cascade is the highest-consequence non-adversarial event in the natural gas sector's Design Basis, precisely because it involves the simultaneous failure of two interdependent sectors. The mechanism operates in both directions: gas supply curtailment causes electric generation shortfalls, which in turn cause compressor stations and control systems that depend on grid electricity to lose power, further degrading gas supply. The cascade can self-sustain even after the initiating event is resolved if the other sector's degraded state blocks recovery actions in one sector.

Adequacy standard: Gas-electric cascade prevention is the primary cross-sector adequacy standard in the natural gas Design Basis. It requires pre-established fuel-supply priority agreements for gas-fired generators critical to grid stability, standby power at critical compressor stations sufficient to sustain operations during grid outages, and coordinated curtailment sequencing protocols between gas transmission operators and electric system operators. These requirements are elaborated in the coordination framework in Section 7.

5. The Military Action Extension

5.1 Why the Non-Adversarial Design Basis Is Insufficient

The non-adversarial Design Basis defined in Section 4 is necessary but insufficient as a complete analytical framework for the natural gas sector. Each bounding event in Section 4 shares a characteristic that distinguishes it from the military-action threat tier: it is not chosen. Equipment fails according to probabilistic distributions. Hurricanes follow meteorological paths. Earthquakes occur along fault lines. The system designer can analyze historical frequencies, model consequence chains, and invest in protection against the events most likely to occur within a defined planning horizon. The adversary does not operate within this framework.

Military action against natural gas infrastructure introduces four characteristics that are absent from the non-adversarial Design Basis and that require an extended analytical framework to address. These four characteristics were identified in the DBA-MA parent framework in the nuclear context; they apply without modification to the natural gas sector.

5.2 The Four Military Action Characteristics

5.2.1 Intentionality

The adversary chooses its targets. It does not attack randomly or according to a probabilistic distribution. It studies the sector, identifies the highest-consequence targets, assesses the defensive posture of those targets, and selects the attack configuration most likely to achieve the desired strategic effect. Intentionality means that the adversary will not attack the hardened compressor station when an equivalent station with weaker physical security is available in the same corridor. It means the adversary will not attack in July, even though an attack in January produces a consequence envelope three times as large. And it means the adversary will not attack the facility the sector has defended when the undefended facility in the same consequence chain produces the same strategic outcome.

The implication for the Design Basis is that the non-adversarial bounding events assume the worst event is determined by physics and probability. The military-action bounding event is determined by adversary choice. The sector must be hardened not only against the probabilistically most likely events but against the strategically most valuable targets, as assessed by a sophisticated adversary who has studied the sector's architecture.

5.2.2 Persistence

Military action does not affect the sector's recovery timeline. A hurricane passes. A compressor failure is repaired when equipment and personnel are available. Military action can be sustained: an adversary that understands the sector's recovery architecture can plan follow-on actions timed to defeat the recovery. The portable compression equipment deployed in response to Scenario A in DBA-GAS-FC1 is itself a target. The supply chain for replacement compressor units is a target. The repair crews mobilizing to restore damaged infrastructure are operating in a threat environment that the non-adversarial bounding events do not contemplate.

The implication for the Design Basis: recovery timelines in the non-adversarial framework assume that repair resources can be mobilized, deployed, and operated without interference. The military-action extension must address recovery under adversary pressure — a fundamentally different planning requirement.

5.2.3 Precision

Modern state actors can execute precision attacks against specific system components. The adversary does not merely threaten the pipeline; it can target the compressor unit at the station with the longest lead time, the storage field with the highest-deliverability buffer, or the LNG terminal train serving the allied supply contract of greatest geopolitical significance. Precision targeting allows the adversary to maximize strategic effect while minimizing the attack's signature, complicating both attribution and response.

The implication for the Design Basis is that the non-adversarial bounding events treat component failures as stochastic. The military-action extension must treat component targeting as strategic, which requires identifying — before the attack — which components produce the highest-consequence failure modes and ensuring that those components receive hardening priority.

5.2.4 Denial

Military action can deny site access. The repair crews, inspection teams, and emergency responders that the non-adversarial recovery framework depends on cannot access a facility under active military threat or designated a hazardous area by ongoing operations. The Zaporizhzhia nuclear facility — Europe’s largest nuclear plant — operated under degraded safety conditions for months precisely because the normal recovery framework — sending in qualified technicians, procuring replacement parts, and implementing emergency procedures — could not function under military occupation. Natural gas compressor stations, storage fields, and pipeline rights-of-way are subject to the same access denial dynamic.

The implication for the Design Basis: protective systems under the military-action extension must incorporate the local sovereignty principle — the requirement that protective actions operate passively, requiring no external power, communications, or operator action to initiate. Recovery architecture must account for delayed access and identify which protective functions can be performed remotely, which require on-site personnel, and the consequences of delayed access.

5.3 The Extended Design Basis: DBA-MA

The DBA-MA extension does not replace the non-adversarial Design Basis — it builds on it. Each non-adversarial bounding event has a military-action analog: the same initiating condition, but deliberately caused, precisely targeted, strategically timed, and executed in a manner designed to defeat the recovery architecture. The extended Design Basis asks: given that the adversary has studied this facility class, identified the highest-consequence targets, and can execute a precision combined cyber-kinetic attack, what is the bounding military-action event, and what does adequate protection against that event require?

The DBA-MA bounding event is in every case more severe than its non-adversarial analog, for four reasons that follow directly from the four characteristics above. First, the adversary selects the highest-consequence configuration; the non-adversarial event occurs according to

probability. Second, the adversary times the attack to maximize consequence; the non-adversarial event is timing-indifferent. Third, the adversary can coordinate simultaneous events that defeat the sector's single-scenario resilience mechanisms; the non-adversarial events are independent. Fourth, the adversary can deny the access on which recovery depends; the non-adversarial recovery framework assumes unimpeded access.

The three design basis scenarios in DBA-GAS-FC1 — the interstate pipeline cascade, the refinery cyber-kinetic sequence, and the combined pipeline rupture and ignition — instantiate the DBA-MA bounding event for the natural gas transmission facility class. The LNG terminal attack use case instantiates it for LNG infrastructure. Subsequent facility-class documents will instantiate it for processing, distribution, and storage.

5.4 The Command-Broker Architecture and the Brightline

The command-broker architecture is the Design Basis requirement for all ML and Gen AI components whose outputs can affect physical process parameters in natural gas sector control systems. The Brightline is the Bright Line as defined in FD-BL (Foundational Definition — The Bright Line): a threshold on individual actions. Under FD-BL §4.2–§4.3, an action on a physical process parameter is above the Brightline when its incorrect performance could produce a consequence at or above the facility's severity threshold, and below it when it could not — placement follows the consequence of the action, not whether the component is labeled advisory or sits in a command layer. The advisory / physical-command-layer seam remains the natural place to build enforcement, but it is not the definition. Above-Brightline actions require a human command broker; the boundary is enforced by architecture, not by procedure or operator discipline. DBA-GAS facility-class instruments inherit this definition by reference to FD-BL and identify, for each facility class, the specific physical process parameters whose actions fall above the Brightline and the command-broker architecture that enforces authorization for them.

The canonical cross-sector instrument governing this architecture is the Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector), published by Eclectic Technologies. That document establishes the broker concept, the Brightline principle, the three-tier criticality framework, and the distinction between the Command Broker and the Influencer Broker. It is the authoritative source for all requirements that this section implements in the natural gas sector context. The CB-Framework derives its three-tier criticality framework from the ISAGCA Supplier Ad-hoc Working Group's analysis of ML and Gen AI risk in critical infrastructure industrial control systems, published as "A Study of AI Risks in Critical Infrastructure -- Vendor Agnostic."

The Three-Tier Criticality Framework. The CB-Framework establishes three tiers based on consequence severity. At the highest criticality tier -- applications whose malfunction could produce the Design Basis consequence thresholds defined in this document: widespread supply curtailment, loss of pipeline pressure integrity, or loss of gas supply to life-safety-dependent loads -- no autonomous Gen AI action is permissible. A human command broker must be in the

control loop for every change to a physical process parameter that a Gen AI component recommends. An ML component that is deterministic within a stated input bound and formally testable as deployed is outside the Command Broker obligation for its in-bound actions under FD-BL §4.4, and may execute those actions autonomously at this tier. That is a carve-out from the scope of the obligation and not a finding that the actions concerned fall below the Bright Line; the Command Broker section below states the distinction and it is not restated here. At mid-level criticality, Gen AI informs operators and operators decide, and an ML component satisfying the same property test may execute its in-bound actions autonomously. At lowest criticality, Gen AI may act as a virtual assistant with limited autonomous action, and the same carve-out governs ML. In every tier, placement remains per action on consequence, inherited by reference from DBA-EN-SF §4.3; the criticality tier sets assessment rigor and does not decide placement. For the natural gas sector Design Basis, the highest-criticality tier governs all applications that control or influence compressor station operations, pipeline pressure management, safety instrumented system parameters, and emergency shutdown logic.

The Command Broker. The command broker is the human operator who decides -- who brokers -- a course of action based on process observables and Gen AI advisory output. The command-broker architecture requires that the control system enforce the broker function mechanically: the system must be incapable of executing a physical process command originating from a Gen AI advisory component without a positive human confirmation step that is local, auditable, and not itself executable remotely or by another software component. An ML component that is deterministic within a stated input bound and formally testable as deployed is outside the Command Broker obligation for its in-bound actions and may execute them autonomously without command-broker interposition; determinism is the scope carve-out, not the placement. Placement is per action on consequence under DBA-EN-SF §4.3, which governs and is not restated here. A vendor characterization of embedded Gen AI as providing "autonomous process optimization" or "direct control integration" describes a capability that is inconsistent with Design Basis compliance in the highest-criticality natural gas sector applications. The vendor characterization does not govern; the Design Basis requirement governs.

Vendor-Embedded ML and Gen AI. Vendor-embedded ML and Gen AI in natural gas sector control systems must be assessed against the Brightline before deployment in any highest-criticality application. The assessment must confirm three things: that the Gen AI component has no pathway to physical command execution without human confirmation; that ML components are deterministic within a stated input bound and formally testable as deployed, and are therefore outside the Command Broker obligation for their in-bound actions under FD-BL §4.4 — a carve-out from scope, not a placement finding; and that Gen AI components have been subjected to red-teaming by parties independent of the vendor, addressing natural gas sector-specific attack scenarios including prompt injection and adversarial inputs designed to produce unsafe compressor or pipeline control outputs.

6. DBA-MA Threat Framework for the Natural Gas Sector

6.1 Threat Actor Taxonomy

The DBA-MA framework defines three threat actor tiers. The Design Basis Threat is drawn from the upper two.

Tier 1 — State Actor. Nation-states with advanced persistent threat capability and the demonstrated willingness to employ combined cyber, kinetic, and influence operations against energy infrastructure. Russia has demonstrated operational execution in Ukraine on a national scale. China has documented pre-positioning activity in U.S. infrastructure networks. Iran has demonstrated its capability against energy-sector targets in the Gulf region. Tier 1 actors have the intelligence resources to conduct multi-year pre-positioning campaigns, the technical capability to execute precision cyber-kinetic attacks, and the strategic patience to time attacks for maximum consequence.

Tier 2 — Near-State Proxy. Paramilitary organizations and hybrid warfare contractors operating under state direction with access to state-provided capabilities. Tier 2 actors provide Tier 1 state actors with plausible deniability for kinetic operations. The relevant scenario involves a Tier 1 state actor directing a Tier 2 proxy to execute kinetic attacks on compressor stations or pipeline facilities. In contrast, the Tier 1 actor maintains cyber pre-positioning and a SCADA compromise that amplifies the physical attack's consequences.

Tier 3 — Sophisticated Non-State Actor. Criminal organizations and ideologically motivated actors with limited state-provided capability or opportunity access. Tier 3 threats are addressed by existing law enforcement and TSA security frameworks. The Colonial Pipeline ransomware incident is a Tier 3 event whose consequences, while significant, were substantially less severe than a Tier 1 or Tier 2 attack would have produced against the same infrastructure.

6.2 Attack Modality Taxonomy

Three primary attack modalities are relevant to the natural gas sector. These modalities are not mutually exclusive; the Design Basis Threat anticipates their coordinated use.

Kinetic Direct. Physical destruction of infrastructure assets using conventional explosives, shaped charges, or incendiary devices. Compressor stations and pipeline rights-of-way are the primary kinetic targets in the natural gas sector. LNG terminal marine infrastructure, storage field wellheads, and processing plant fired equipment are facility-class-specific kinetic targets addressed in their respective documents.

Cyber-Kinetic Combined Arms. Integration of ICS/SCADA compromises with kinetic action. The cyber component serves to suppress protective systems before the kinetic strike, amplify physical damage by manipulating process conditions at the moment of attack, delay detection and attribution, and complicate recovery by corrupting control system state. The design basis

threat combines Tier 1 cyber pre-positioning — potentially months before the kinetic strike — with Tier 2 kinetic execution, exploiting the gap between a state actor's sophisticated cyber capabilities and the plausible deniability of a proxy kinetic operation.

Logistics and Supply Chain Denial. Interdiction of the supply chains for critical replacement equipment — compressor turbines, pipeline segments, processing plant catalysts, specialty cryogenic equipment for LNG facilities — to extend recovery timelines after a kinetic attack. A Tier 1 adversary that understands compressor unit replacement lead times (12 to 24 months) can plan supply chain interdiction operations that sustain disruption long after the physical damage from the kinetic attack has been assessed, and repair has begun.

6.3 Bounding Logic

The Design Basis Threat is not the most severe conceivable attack. It is the attack that sector hardening and resilience architecture must withstand while maintaining critical functions. The DBA-MA bounding parameters for the natural gas sector are: Tier 1 cyber pre-positioning with Tier 2 kinetic execution; a combined cyber-kinetic modality; target selection for maximum cascade effect across the gas-electricity interdependency; timing at peak demand, coincident with degraded emergency response capacity; and simultaneity within a window too short for adaptive response.

6.4 Simultaneity as a Design Parameter

The natural gas sector’s existing resilience mechanisms — storage drawdown, mutual assistance, alternative routing, fuel switching — are designed for sequential recovery: one event occurs, restoration resources mobilize, and the system recovers before the next challenge. The Design Basis Threat violates this assumption. A sophisticated adversary will execute coordinated attacks on multiple nodes within a time window that exceeds the sector’s adaptive capacity, using the response to the first attack to position for the second, and targeting the restoration resources that would address the first attack as elements of the second.

Simultaneity is therefore a design parameter, not merely an operational risk. Security investments that improve single-node resilience provide limited benefit against the Design Basis Threat if they do not also address the coordination and resource pre-positioning requirements for simultaneous multi-node events. This is the basis for the “simultaneity trap” analysis in DBA-GAS-FC1 Section 5.2.

7. Consequence Interaction and Cross-Sector Dependencies

7.1 The Gas-Electricity Interdependency Loop

The gas-electricity interdependency is the most consequential cross-sector relationship in the natural gas sector's consequence analysis. It operates in both directions and, under the right conditions, can produce a self-sustaining feedback loop that exceeds either sector's independent recovery capability.

The forward direction: natural gas supply curtailment reduces the fuel available to gas-fired electric generators. Those generators reduce output or shut down. The resulting generation shortfall degrades grid reliability and, if it exceeds the available reserve margin, triggers load shedding or cascading outages.

The reverse direction: degradation of electric grid reliability affects gas transmission operations. Compressor stations at many facilities depend on grid-supplied electricity for auxiliary systems, instrumentation, and control. Storage field pumping and monitoring systems depend on grid power. Pipeline control centers require reliable electricity for SCADA operations. A grid reliability event triggered by gas curtailment can therefore further degrade gas transmission capability, completing the feedback loop.

Under military-action conditions, this interdependency becomes a force multiplier for the adversary. A Tier 1 actor that understands the interdependency will not attack only the gas sector or only the electricity sector. It will attack the seam between them — the facilities and operational relationships where the two sectors' vulnerabilities compound. The compressor stations that have no backup power, the gas-fired generators on interruptible contracts that lose fuel first, the SCADA centers whose UPS systems were not designed for extended outages: these are the nodes where the feedback loop initiates.

7.2 Cascade Logic and Multi-Sector Consequence Planning

Three categories of cascades are relevant to military-action consequence planning for the natural gas sector.

Gas-to-electric cascade. Gas supply curtailment → generation shortfall → grid reliability degradation → compressor station power loss → further gas supply curtailment. This is the primary cascade path and the one for which pre-planned coordination protocols are most urgently needed.

Gas-to-heating cascade. Gas supply curtailment → LDC curtailment of residential and commercial customers → loss of space heating → hypothermia risk for vulnerable populations → hospital system surge → emergency medical services overload. This cascade is most consequential during winter heating-season events and drives the life-safety classification of major gas supply disruptions.

Gas-to-logistics cascade. Gas supply curtailment → curtailment of industrial customers, including petroleum refineries and chemical plants → reduced fuel production and industrial output → supply chain disruption → military logistics and emergency response resource

degradation. This cascade path is slower than the gas-to-electric path but has longer-duration consequences.

Multi-sector consequence planning for gas-electric cascade events requires pre-established protocols among natural gas transmission operators, electric utilities, regional transmission organizations, and federal coordination bodies. The coordination framework must address curtailment sequencing that protects gas supply to the generators most critical to grid stability, backup power requirements at the compressor stations most critical to gas supply continuity, and communication and decision authority protocols for the first 24 to 48 hours of a cascading event when the consequences are still responsive to intervention.

7.3 Seasonal Amplification

The consequence envelope for military-action events against the natural gas sector is materially larger in winter than at other times of year. Winter demand for natural gas peaks simultaneously for residential and commercial space heating and for electric power generation fuel supply. Storage inventory, while at its annual high entering the winter season, depletes fastest precisely when it is most needed as a supply buffer. An adversary planning a Scenario A or C attack against Northeast or Midwest natural gas infrastructure in January is not executing the same attack as in July — the consequence envelope is qualitatively different in scale and in the severity of life-safety consequences.

Seasonal amplification is an explicit design parameter in the DBA-MA framework. Security posture, protective resources, and emergency response pre-positioning should be adjusted seasonally to reflect the larger consequence envelope during peak demand periods. The adequacy standards for recovery capability in Section 3.4 must be met under peak-demand conditions, not average-demand conditions.

8. Protective Action and Recovery Architecture

8.1 Threat Detection Framework

The DBA-MA framework distinguishes two indicator categories whose recognition drives protective action decisions.

Elevated threat indicators raise the probability that a military action event is being planned or prepared, without confirming an imminent attack. These include intelligence community reporting of adversary interest in natural gas infrastructure or compressor station reconnaissance; anomalous ICS network activity, including unusual read activity on process historian data or lateral movement between IT and OT network segments; physical surveillance of compressor stations or pipeline facilities; and unusual acquisition patterns for commercially available explosives, pipeline mapping data, or process engineering documentation.

Imminent threat indicators suggest that an attack is in preparation or has been initiated. These include confirmed physical intrusion at pipeline or compressor station facilities; anomalous pressure readings at multiple pipeline nodes within a single operational system within a short time window; loss of communication with SCADA systems at multiple locations within a single corridor simultaneously; and control system parameter changes inconsistent with operator-initiated actions, particularly affecting safety instrumented system parameters.

8.2 Decision Authority Framework

Protective action decisions require clear authority at each level of the response hierarchy. Facility operators retain authority to initiate emergency shutdown, segment isolation, and site evacuation without external authorization. Sector-level coordination through the ONE-ISAC and the O&G Subsector Coordinating Council facilitates mutual assistance and ensures TSA and CISA notification. Federal ESF-12 activation under DOE leadership coordinates federal support for supply restoration. Strategic Petroleum Reserve activation for petroleum product components of the consequence chain requires the Secretary of Energy or Presidential authority, as appropriate.

8.3 Federal Coordination Triggers

The following events constitute Federal Coordination Triggers — thresholds that require mandatory federal coordination and, where authority exists, federal response action: confirmed coordinated attack on two or more pipeline or compressor station facilities within a single operational system within 24 hours; natural gas supply disruption affecting electric power generation in a region with less than 72 hours of alternative fuel capacity; ICS compromise with confirmed or strongly suspected nation-state attribution at two or more sector facilities; confirmed fatalities at pipeline or compressor station facilities in circumstances consistent with adversary action.

8.4 Recovery Sequencing Principles

Recovery from a military-action event against the natural gas sector follows a phased sequence that reflects the prioritization of life-safety functions, the operational constraints on restoration under adversary pressure, and the resource limitations of simultaneous multi-node events.

Phase 1 — Immediate stabilization. Segment isolation to contain physical damage. Storage drawdown activation at maximum deliverability rates to sustain priority customer supply. LDC curtailment protocol activation under defined priority sequencing: interruptible industrial and commercial customers first; firm-service customers maintained until supply is insufficient to serve them; life-safety infrastructure protected throughout—gas-electric coordination to activate the generator fuel supply for grid-critical units.

Phase 2 — Short-term bridging. Alternative supply routing activation. Portable compression deployment where terrain and system configuration permit. Canadian import maximization on

existing interconnections. LDC peak-shaving LNG facility activation, where available. Federal ESF-12 coordination for supply restoration support.

Phase 3 — Medium-term restoration. Compressor station structural repair and equipment procurement. Pipeline segment repair at rupture sites. Regulatory inspection and return-to-service process. Incremental load restoration in reverse-curtailment order. Supply chain monitoring for adversary interdiction of replacement equipment.

The local sovereignty principle applies to Phase 1 protective actions. Isolation valves, emergency shutdown systems, and safety instrumented systems must be capable of executing their protective functions without external power, communications, or operator intervention. Phase 2 and Phase 3 actions require the deployment of external resources and are therefore subject to the access-denial constraint under military-action conditions.

9. Facility Class Register and Inheritance Model

9.1 The Inheritance Model

Facility-class documents in the DBA-GAS series inherit the sector framework's threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards. They do not repeat this material. Each facility-class document adds what is specific to its asset profile: facility-class-specific bounding events, consequence chains, design parameters, and recovery architecture, which the sector framework establishes in general terms but does not develop for each asset class.

Each facility-class instrument must include a section explicitly identifying which elements are inherited from this document unchanged, which elements are inherited with facility-class-specific adaptation, and which elements are entirely locally defined. This inheritance map is a required component of every facility-class instrument at the time of first publication. It must be updated at each revision to reflect any changes to this sector framework.

Use cases issued under the sector framework apply the DBA-MA methodology to a specific attack vector or consequence chain within a facility class's scope. Policy companions outline the regulatory, legislative, or institutional actions required to address identified gaps. This modular architecture allows the series to grow as new facility classes are analyzed and new use cases are developed, without requiring revision of the foundational framework documents.

9.2 Facility Class Placement Rationale

The natural gas sector is organized around the physical pathway from wellhead to end-use burner tip. Facility classes in the DBA-GAS series are defined by their position in that pathway and by the consequence chain their loss initiates. Transmission and compression infrastructure (FC1) is placed first because compressor station outage is the highest-consequence single-facility event in

the gas sector: it disrupts the pressure differential that moves gas across hundreds of miles of pipeline. Processing infrastructure (FC2) is placed second because processing plant outage curtails wellhead deliverability upstream of the transmission system, reducing the gas available to compress and transport. Underground storage (FC3) addresses the demand-buffering layer that absorbs seasonal imbalance; its loss is most consequential during peak winter heating demand. Local distribution (FC4) addresses the last-mile delivery system; its loss directly interrupts supply to residential and commercial end-users. LNG terminals (FC-LNG) are placed last as a promoted facility class. Originally addressed as a use case for the consequence envelope of a military-action-induced LNG release, they were elevated to a standing facility class on the determination that LNG export infrastructure carries a strategic significance — allied supply commitments and a distinct marine attack surface — that warrants a recurring design basis rather than a single-scenario treatment. The use-case instrument DBA-GAS-UC-LNG is retained as the facility class's bounding-event annex.

9.3 Facility Class Register

The following table is the current facility class register for the DBA-GAS series. Facility classes designated as Anticipated require specialist development by qualified natural gas sector engineers and safety analysts. Their inclusion in the register establishes the planned scope of the series; it does not represent completed analytical work.

Designation	Facility Class	NIPP Sector	Status	Primary Regulatory Anchor
DBA-GAS-FC1	Natural gas transmission -- interstate pipeline and compressor stations	Energy	Active	PHMSA 49 CFR Part 192; TSA Pipeline Security Directives; FERC Gas Tariff
DBA-GAS-FC2	Natural gas processing	Energy	Active	PHMSA 49 CFR Part 192; EPA RMP (40 CFR Part 68); OSHA PSM (29 CFR 1910.119)
DBA-GAS-FC3	Underground storage -- depleted reservoir and salt cavern	Energy	Active	PHMSA 49 CFR Part 192 Subpart U; FERC; state conservation agency jurisdiction
DBA-GAS-FC4	Local distribution -- LDC city gate to end-use	Energy	Active	PHMSA 49 CFR Part 192; state PUC jurisdiction
DBA-GAS-FC-LNG	LNG terminals -- liquefaction, storage, and marine transfer (promoted from DBA-GAS-UC-LNG)	Energy	Active -- promoted; annex DBA-GAS-UC-LNG	PHMSA 49 CFR Part 193; USCG 33 CFR Part 127; FERC NGA §3 siting

9.4 Current Use Cases and Policy Companions

The following use case and policy companion documents are currently issued under this sector framework.

Designation	Document	Status
DBA-GAS-UC-LNG	LNG Terminal Attack	Active (bounding-event annex to DBA-GAS-FC-LNG)
DBA-GAS-POL	Deterrence Without Standards	Active

9.5 Development Sequence

The four pathway facility classes are issued: FC1 (Natural Gas Transmission), FC2 (Natural Gas Processing), FC3 (Underground Storage) and FC4 (Local Distribution). FC2's consequence chain is distinct from the transmission case and required specialist process safety analysis; FC3 addresses the demand-buffering role that distinguishes the gas sector's resilience architecture from electric sector analogs; and FC4 required engagement with local distribution company operators, whose regulatory posture and operational architecture differ materially from interstate transmission operators. The BLEVE, vapor cloud explosion, and thermal radiation consequence envelope for LNG releases is being incorporated into DBA-GAS-UC-LNG as an expansion rather than a separate document.

10. Gaps and Open Items

10.1 Regulatory Gap: Absence of Mandatory Physical Security Standards

The sector framework’s single most significant finding is the absence of mandatory physical security standards for the natural gas transmission sector. This gap is fully addressed in DBA-GAS-POL. Its relevance to the sector framework is this: the Design Basis established in this document has no regulatory mechanism to compel its implementation. A sector framework that defines bounding events and adequacy standards without a compliance mechanism is an analytical contribution, not a protective one. Translating the Design Basis into enforceable standards — through a process modeled on the NERC CIP-014 development — is the single most consequential action available to close the gap this framework identifies.

10.2 Data Gap: Facility-Level Design Basis Inventories

The non-adversarial Design Basis in Section 4 establishes the sector-level bounding event framework. It does not establish facility-level design parameters: the specific equipment ratings, site configurations, and operational constraints that determine whether a given facility meets the adequacy standards defined here. That inventory does not exist in a form accessible to the sector-level analysis. Developing it requires engagement with facility operators and a data-collection framework that currently lacks a regulatory basis. The mandatory standards process — which

would include third-party assessment requirements analogous to CIP-014's independent verification — is the mechanism through which the facility-level inventory becomes available.

10.3 Coordination Gap: Gas-Electric Pre-Planning

The gas-electricity cascade analysis in Section 7 identifies the interdependency loop as the highest-consequence cross-sector vulnerability in the natural gas Design Basis. Pre-established coordination protocols among gas operators, electric utilities, and RTO/ISOs for the cascade scenario are inadequate. The NERC Electricity Subsector Coordinating Council and the O&G Subsector Coordinating Council have not jointly developed the operational protocols — specific curtailment sequencing, backup power requirements, communication authority — that the cascade scenario demands. This is an open item that does not require legislation to address; it requires coordination and exercise activity among the parties.

10.4 Conferred Reference Events

On the author's determination of 14 October 2026 (*Energy and Nuclear Reference Events Follow the Same Method*, held in Project 0), this framework confers reference events for the natural gas sector by the same method as every other sector framework in the corpus. A facility class of this series proposes a real event; this section confers it and assigns a designator of the form DBA-GAS-RE-N, numbered in the order of conferral; Project 0 enters it in the DBA Reference Event Register; and Project 11, DBA Reference Events, writes the instrument that reads the event back through the proposing class's design basis. A designator commits the series to a register row and to a standalone instrument, and the instrument's recommendations return to the proposing class.

An event already conferred in another sector is cited here by its designator and is not conferred again. Among them is DBA-COMM-RE-1 (*Baltic Sea Cable Incidents*), whose reading covers the damage to the Balticconnector gas pipeline in the same seabed corridor.

On the author's decision of 15 October 2026, this framework confers one event. DBA-GAS-RE-1 is the uncontrolled release of natural gas from well SS-25 at the Southern California Gas Company's Aliso Canyon storage field, Los Angeles County, California, from 23 October 2015 until the well was sealed on 18 February 2016. It is proposed by Underground Storage (DBA-GAS-FC3) at its v0.8, Section 8.6, where Aliso Canyon is also the anchor event of postulated event EF-1 and of the class's return-to-service constraint. The primary sector is Energy, natural gas sub-sector. The impacted sectors the class proposes, for the instrument to confirm, are Energy, electric sub-sector, through the risk to gas-fired generation in the Los Angeles Basin assessed in 2016, and Healthcare and Public Health, through the temporary relocation of households beside the field. The tier reading held here is provisional and the instrument's to determine: a non-adversarial well-integrity failure, with none of the four characteristics present,

realizing EF-1 in full. The regulators' and root-cause reports are primary, and the instrument makes no finding of fault.

The designator is numbered in the order of conferral; the number records conferral, not issue. It is sector-primary and single. Project 0 enters it in the DBA Reference Event Register, and Project 11 has written the instrument, issued on 1 November 2026; it confirms the provisional non-adversarial reading.

No other event is conferred here. The other real events this framework and its classes name as evidence, among them Hurricanes Katrina and Rita and Winter Storm Uri, remain anchor events for the postulated events they support. An anchor becomes a reference event only when a class proposes it and this section confers it.

Revision History

Version	Date	Author	Description
1.0	April 2026	Tim Roxey	Initial version. Establishes the Design Basis and DBA-MA framework for the natural gas sector. Introduces the non-adversarial Design Basis methodology and bounding event categories. Defines the four characteristics of military action and their implications. Establishes the sector-level threat taxonomy, consequence interaction framework, and facility-class instrument architecture.
1.1	April 2026	Tim Roxey	Section 1.3 and Section 9.2: DBA-GAS-UC-LNG version updated from v1.0 (expansion in progress) to v1.1 (complete). Section 9.2 description updated to reflect incorporation of BLEVE, VCE, and pool fire physical consequence analysis.
1.2	May 2026	Tim Roxey	Section 4 opening paragraph updated to carry an explicit three-layer (DB → DBA → DBA-MA) sequence statement.
1.3	May 2026	Tim Roxey	Section 9 — "Facility Class Instruments" replaced with full "Facility Class Register and Inheritance Model" architecture (Sections 9.1–9.5). Existing inheritance model prose retained and extended with a mandatory inheritance-map requirement. New Sections 9.2 (Placement Rationale), 9.3 (five-column register table), 9.4 (Current Use Cases and Policy Companions table), and 9.5 (Development Sequence) added. Addresses Gap 2.2 of DBA Series Gaps and Recommendations v1.2.
1.5	June 2026	T. Roxey	Housekeeping revision. FC2 register entry updated from Anticipated to Active -- v0.1 (DBA-GAS-FC2: Natural Gas Processing). FC3 register entry updated from Anticipated to Active -- v0.1 (DBA-GAS-FC3: Underground Storage -- Depleted Reservoir and Salt Cavern). No substantive content changes.
1.6	June 2026	T. Roxey	Housekeeping revision. FC4 register entry updated from Anticipated to Active -- v0.1 (DBA-GAS-FC4: Local Distribution -- LDC City Gate to End-Use). All four DBA-GAS facility class instruments are now at first issue. No substantive content changes.
1.7	June 2026	T. Roxey	Marine/strategic terminal classification. LNG terminals promoted from use case to a standing facility class, registered as DBA-GAS-FC-LNG in §9.3 with strategic-significance rationale (§9.2). DBA-GAS-UC-LNG dispositioned as the bounding-event annex to DBA-GAS-FC-LNG (§9.4). Promotion asserted in DBA-GAS-MA-SF v0.2 and now registered here. Parallels the petroleum FC3 split in DBA-OIL-SF; register the two together at the parent. No other substantive change.

1.8	September 2026	T. Roxey	§ AI-governance placements conformed under OI-R3-1 (Governing Instructions v0.17 §6) on the inherit-by-reference model. Placements previously asserted from the character of the component — deterministic, ML, or protective — are restated as the FD-LD scope carve-out applied to in-bound actions, with placement itself referred to DBA-EN-SF §4.2 rather than restated in this document. Restating the placement criterion per instrument is the drift mechanism that allowed the error to propagate; this revision removes the restatement. No physical Design Basis content, consequence threshold, event code, scenario, or metric is changed.
1.9	September 2026	T. Roxey	CR-VA conformance. The OI-R3-1 pass recorded at v1.8 conformed this document's statement of the construct and left three of its applications standing; this revision conforms the applications and changes no physical Design Basis content. Three-Tier Criticality Framework: the highest-tier sentence granting ML autonomy because its behavior is bounded and verifiable, and concluding that it does not require a command broker, is replaced by the FD-BL §4.4 scope carve-out — a carve-out from the obligation and not a finding that the actions fall below the Bright Line — deferring to the Command Broker section rather than restating it; the mid-level and lowest-tier sentences are conformed to the same reading; and a closing sentence states that placement remains per action on consequence, inherited by reference from DBA-EN-SF §4.3, the criticality tier setting assessment rigor rather than placement. Vendor-Embedded ML and Gen AI: the assessment criterion reading deterministic and formally testable and may therefore operate autonomously is conformed to the same carve-out. The live placement reference is repaired from DBA-EN-SF §4.2 to §4.3, §4.2 being Dual Design Basis Architecture and §4.3 the placement section since DBA-EN-SF v0.15; the v1.8 entry naming §4.2 is a record and stays as issued. CB-Framework_Cross_Sector v0.1 is stripped under the family citation form, CB-REG being authoritative. Raised by Project 2's determination of 8 September 2026 that the autonomy prohibition is keyed to maloperation rather than to a criticality band, reaching this project through Projects 2 and 9 on 9 September. Three successive screens under-reported the site count here: the figure of four is a reading, not a screen result.
1.10	October 2026	T. Roxey	Section 10.4 (Conferred Reference Events) added under the determination of 14 October 2026, Energy and Nuclear Reference Events Follow the Same Method. No event is yet conferred. The events named as evidence for bounding cases in Section 4 are now called anchor events, so that “reference event” keeps its corpus meaning. No section is renumbered.
1.11	October 2026	T. Roxey	Section 10.4 confers DBA-GAS-RE-1, the Aliso Canyon release of October 2015 to February 2016, proposed by DBA-GAS-FC3 (Underground Storage) at v0.8 §8.6, on the author’s decision of 15 October 2026. The section states the event, the proposing class, the primary sector (Energy, natural gas sub-sector), the impacted sectors the class proposes and a provisional tier reading. The instrument is owed by Project 11. No other section is renumbered or changed.
1.12	November 2026	T. Roxey	Citation conformance ahead of the CIKR example-site edition, on the author's approval of 1 November 2026 of Project 0's sign-off request of 31 October 2026. The DBA-MA Parent Framework pin on the cover and in §1.3 is brought from IAEA L1 v1.4 to v2.3, the current master, and is not stripped, Project 6 having issued no register. The §1.3 inventory sentence is stripped of its version tokens and completed with FC2, FC3 and FC4, which it had omitted. Version tokens are stripped from the Status cells of the §9.3 Facility Class Register and the §9.4 table, version state being DBA-EN-REG's. §9.5 no longer describes FC2, FC3 and FC4 as yet to be developed. §1.3 gains the series and Command Broker family citation lead-ins, the series lead-in in its conformed form naming DBA-EN-REG; this framework has no references section. The record of

			proposal in §10.4 and the housekeeping records in this history stand as issued. No design basis content is changed.
1.13	November 2026	T. Roxey	§10.4 records that Project 11 issued the DBA-GAS-RE-1 instrument on 1 November 2026, confirming the provisional non-adversarial reading, where v1.12 said it was not yet written. Issued with DBA-GAS-FC3 v0.10, which takes in the instrument's Erratum 1, ahead of the CIKR example-site freeze. The instrument's finding R-4, noted to this framework, is held for the author's discussion. No other text is changed.